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Vehicular heat management system

Abstract

A vehicular heat management system includes a compressor arranged on a heat pump type refrigerant circulation line, a high-pressure side heat exchanger arranged on the heat pump type refrigerant circulation line, an outdoor heat exchanger arranged on the heat pump type refrigerant circulation line, a plurality of expansion valves arranged on the heat pump type refrigerant circulation line, a low-pressure side heat exchanger arranged on the heat pump type refrigerant circulation line, a first expansion valve arranged on the upstream side of the outdoor heat exchanger, a second expansion valve arranged on the downstream side of the outdoor heat exchanger, and a control part configured to control opening degrees of the first expansion valve and the second expansion valve depending on whether icing occurs in the outdoor heat exchanger under a heat pump mode condition.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This patent application is a national phase under 35 U.S.C. § 371 of International Patent Application No. PCT/KR2022/010263 filed Jul. 14, 2022, which claims the benefit of priority from Korean Patent Application No. 10-2021-0096790 filed Jul. 23, 2021, each of which is hereby incorporated herein by reference in its entirety for all purposes.

TECHNICAL FIELD

(2) The present invention relates to a vehicular heat management system and, more particularly, to a vehicular heat management system capable of coping with the occurrence of icing in an outdoor heat exchanger without restricting the flow of a refrigerant to the outdoor heat exchanger when the icing occurs in the outdoor heat exchanger under a heat pump mode condition and capable of preventing the instability of a refrigerant flow in a refrigerant circulation line due to the restriction of a refrigerant flow to the outdoor heat exchanger and the resultant deterioration of heating performance in the vehicle interior.

BACKGROUND ART

(3) Examples of an eco-friendly vehicle include an electric vehicle, a hybrid vehicle, and a fuel cell vehicle (hereinafter collectively referred to as “vehicle”).

(4) Such a vehicle is equipped with various heat management devices. For example, as shown in FIG. 1, the heat management devices include an air conditioner **10** for cooling and heating a vehicle interior, and a water-cooled cooling device **20** for cooling a battery B, and an electric component module P.

(5) The air conditioner **10** is of a heat pump type and is provided with a refrigerant circulation line **12**.

(6) The refrigerant circulation line **12** includes a compressor **12a**, a high-pressure side indoor heat exchanger **12b**, a first expansion valve **12c**, an outdoor heat exchanger **12d**, a second expansion valve **12e**, a third expansion valve **12f** parallel to the second expansion valve **12e**, a chiller **12g** arranged on the second expansion valve **12e**, and a low-pressure side indoor heat exchanger **12h** arranged on the downstream side of the third expansion valve **12f**.

(7) In the refrigerant circulation line **12**, the first expansion valve **12c** is opened in an air conditioner mode, so that the refrigerant in the compressor **12a** can circulate in the order of the high-pressure side indoor heat exchanger **12b**, the outdoor heat exchanger **12d**, the second expansion valve **12e**, the third expansion valve **12f**, the chiller **12g**, and the low-pressure side indoor heat exchanger **12h**.

(8) Through this refrigerant circulation, a low-temperature cold air is generated in the chiller **12g** and the low-pressure side indoor heat exchanger **12h**. The generated cold air is delivered to the vehicle interior, the battery B, and the electric component module P, thereby cooling the vehicle interior, the battery B, and the electric component module P.

(9) Further, in a heat pump mode, the first expansion valve **12c** is turned on, so that the refrigerant

in the compressor **12a** circulates in the order of the high-pressure side indoor heat exchanger **12b**, the first expansion valve **12c**, and the outdoor heat exchanger **12d**.

(10) In addition, high-temperature heat is generated in the high-pressure side indoor heat exchanger **12b** through the circulation of the refrigerant. The generated heat is supplied to the vehicle interior to heat the vehicle interior.

(11) The refrigerant circulation line **12** further includes an electric component waste heat chiller **12i**. The electric component waste heat chiller **12i** is configured to allow the refrigerant in the refrigerant circulation line **12** returning from the outdoor heat exchanger **12d** to the compressor **12a** to exchange heat with the cooling water in a cooling water circulation line **22** of a water-cooled cooling device **20** that absorbs the waste heat from the battery B and the electric component module P.

(12) Therefore, the waste heat of the battery B and the electric component module P can be recovered to the refrigerant of the refrigerant circulation line **12**. The heat pump mode efficiency of the air conditioner **10** can be increased through this waste heat recovery.

(13) Meanwhile, when the temperature of the outdoor heat exchanger **12d** is lowered to a low temperature under a specific external air condition, for example, a high humidity condition, while the air conditioner **10** enters the heat pump mode, there is a concern that icing may occur on the surface of the outdoor heat exchanger **12d**.

(14) Thus, the air conditioner **10** further includes an icing prevention part **14** for preventing the occurrence of icing in the outdoor heat exchanger **12d**.

(15) The icing prevention part **14** includes a three-way flow control valve **14a** that bypasses the refrigerant before being introduced into the outdoor heat exchanger **12d** to the electric component waste heat chiller **12i**.

(16) When icing occurs in the outdoor heat exchanger **12d**, the icing prevention part **14** restricts the flow of the refrigerant to the outdoor heat exchanger **12d** by bypassing the refrigerant before being introduced into the outdoor heat exchanger **12d**, thereby preventing the icing of the outdoor heat exchanger **12d**.

(17) However, such a conventional heat management system needs to change the flow of the refrigerant in the refrigerant circulation line **12** through the three-way flow control valve **14a** when icing occurs in the outdoor heat exchanger **12d** while entering the heat pump mode. Therefore, the flow of the refrigerant becomes unstable, and the efficiency of the heat pump mode is lowered.

(18) In particular, the refrigerant flow in the refrigerant circulation line **12** is frequently changed while the three-way flow control valve **14a** is repeatedly operated to prevent icing of the outdoor heat exchanger **12d**. Due to this frequent change in the refrigerant flow, the refrigerant flow in the circulation line **12** becomes unstable. The heat exchange rate in the high-pressure side indoor heat exchanger **12b** fluctuates and the temperature change of the air discharged into the vehicle interior intensifies. As a result, the heating performance in the vehicle interior is lowered.

(19) In addition, the conventional heat management system has a structure that restricts the flow of the refrigerant to the outdoor heat exchanger **12d** when icing occurs in the outdoor heat exchanger **12d**. Therefore, the heat exchange efficiency (evaporation efficiency) of the refrigerant is relatively lowered. Thus, the amount of heat generated in the high-pressure side indoor heat exchanger **12b** is also relatively reduced, resulting in poor heating performance in the vehicle interior.

SUMMARY

(20) In view of the problems inherent in the related art, it is an object of the present invention to provide a vehicular heat management system capable of coping with the occurrence of icing in an outdoor heat exchanger without restricting the flow of a refrigerant to the outdoor heat exchanger when the icing occurs in the outdoor heat exchanger.

(21) Another object of the present invention is to provide a vehicular heat management system capable of preventing the instability of a refrigerant flow in a refrigerant circulation line due to the restriction of a refrigerant flow to the outdoor heat exchanger when icing occurs in the outdoor heat

exchanger.

(22) A further object of the present invention is to provide a vehicular heat management system capable of preventing a change in a heat exchange rate in a high-pressure side indoor heat exchanger due to unstable refrigerant flow in a refrigerant circulation line and a change in a temperature of an air discharged into the vehicle interior, thereby improving the heating performance in the vehicle interior.

(23) A still further object of the present invention is to provide a vehicular heat management system capable of preventing the reduction of refrigerant heat exchange efficiency (evaporation efficiency) on the outdoor heat exchanger side due to the restriction of the refrigerant flow to the outdoor heat exchanger.

(24) A yet still further object of the present invention is to provide a vehicular heat management system capable of preventing a decrease in a heat generation amount in a high-pressure side indoor heat exchanger, thereby improving the heating performance in the vehicle interior.

(25) In order to achieve these objects, there is provided a vehicular heat management system, including: a compressor arranged on a heat pump type refrigerant circulation line; a high-pressure side heat exchanger arranged on the heat pump type refrigerant circulation line; an outdoor heat exchanger arranged on the heat pump type refrigerant circulation line; a plurality of expansion valves arranged on the heat pump type refrigerant circulation line; a low-pressure side heat exchanger arranged on the heat pump type refrigerant circulation line; a first expansion valve arranged on the upstream side of the outdoor heat exchanger; a second expansion valve arranged on the downstream side of the outdoor heat exchanger; and a control part configured to control opening degrees of the first expansion valve and the second expansion valve depending on whether icing occurs in the outdoor heat exchanger under a heat pump mode condition.

(26) In the system, the control part may be configured to enter a dual expansion mode in which a refrigerant is expanded by the first expansion valve and the second expansion valve if the control part determines that icing occurs in the outdoor heat exchanger, and enter a single expansion mode in which a refrigerant is expanded by only one of the first expansion valve and the second expansion valve if the control part determines that icing does not occur in the outdoor heat exchanger.

(27) In the system, the control part may be configured to, in the single expansion mode, fully open the second expansion valve on the downstream side of the outdoor heat exchanger and control the opening degree of the first expansion valve on the upstream side of the outdoor heat exchanger to adjust a refrigerant flow rate on the side of the outdoor heat exchanger.

(28) In the system, the control part may be configured to, in the dual expansion mode, control the opening degrees of the first expansion valve and the second expansion valve to adjust the refrigerant flow rate on the side of the outdoor heat exchanger.

(29) In the system, the control part may be configured to, in the dual expansion mode, control the opening degrees of the first expansion valve and the second expansion valve to adjust the refrigerant flow rate on the side of the outdoor heat exchanger in a decreasing direction so that the temperature of the outdoor heat exchanger is increased to cope with occurrence of icing.

(30) In the system, the control part may be configured to, in the dual expansion mode, keep the opening degree of one of the first expansion valve and the second expansion valve constant, and variably control the opening degree of the other of the first expansion valve and the second expansion valve.

(31) In the system, the control part may be configured to, in the dual expansion mode, keep the opening degree of the first expansion valve constant, and variably control the opening degree of the second expansion valve.

(32) In the system, the control part may be configured to, in the dual expansion mode, periodically predict the probability of occurrence of icing in the outdoor heat exchanger according to a pre-stored logic, and variably control the opening degree of the second expansion valve according to

the predicted probability of occurrence of icing.

(33) In the system, the control part may be configured to, in the dual expansion mode, periodically calculate an icing occurrence index of the outdoor heat exchanger at predetermined time intervals through pre-stored Formula 2: icing occurrence index (I)=outdoor air temperature (Tamb)–outdoor heat exchanger outlet refrigerant temperature (Tref), the control part may be configured to predict that the probability of occurrence of icing in the outdoor heat exchanger is high if the calculated icing occurrence index is greater than a pre-stored reference index, and the control part may be configured to predict that the probability of occurrence of icing in the outdoor heat exchanger is low if the calculated icing occurrence index is smaller than the reference index.

(34) According to the vehicular heat management system of the present invention, when icing occurs in the outdoor heat exchanger under a heat pump mode condition, the temperature of the outdoor heat exchanger is adjusted by controlling the opening degree of the third expansion valve on the downstream side of the outdoor heat exchanger. Therefore, when icing occurs in the outdoor heat exchanger, it is possible to cope with the occurrence of icing in the outdoor heat exchanger without restricting the flow of the refrigerant to the outdoor heat exchanger.

(35) In addition, since it is possible to cope with the occurrence of icing in the outdoor heat exchanger without restricting the flow of the refrigerant to the outdoor heat exchanger, the unstable flow of the refrigerant in the refrigerant circulation line due to the restriction of the flow of the refrigerant to the outdoor heat exchanger can be prevented when icing occurs in the outdoor heat exchanger.

(36) In addition, since the instability of the flow of the refrigerant in the refrigerant circulation line due to the restriction of the flow of the refrigerant to the outdoor heat exchanger can be prevented, it is possible to prevent the fluctuation of the heat exchange rate in the high-pressure side indoor heat exchanger due to the instability of the flow of the refrigerant in the refrigerant circulation line and the resultant change in the temperature of the air discharged into the vehicle interior, thereby improving the heating performance in the vehicle interior.

(37) In addition, since it is possible to cope with the occurrence of icing in the outdoor heat exchanger without restricting the flow of the refrigerant to the outdoor heat exchanger when icing occurs in the outdoor heat exchanger, the decrease in the refrigerant heat exchange efficiency (evaporation efficiency) on the outdoor heat exchanger side due to the restriction of the refrigerant flow to the outdoor heat exchanger can be prevented.

(38) In addition, since it is possible to prevent the decrease in the refrigerant heat exchange efficiency (evaporation efficiency) on the outdoor heat exchanger side due to the restriction of the refrigerant flow to the outdoor heat exchanger, it is possible to prevent a decrease in the heat generation amount in the high-pressure side indoor heat exchanger, thereby improving the heating performance in the vehicle interior.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram showing a conventional vehicular heat management system.

(2) FIG. 2 is a view showing a vehicular heat management system according to a first embodiment of the present invention.

(3) FIG. 3 is a p-h diagram for explaining an operation example of the vehicular heat management system according to the first embodiment of the present invention, showing the control of the vehicular heat management system when no icing occurs in an outdoor heat exchanger in a heat pump mode.

(4) FIG. 4 is a p-h diagram for explaining an operation example of the vehicular heat management system according to the first embodiment of the present invention, showing the control of the

vehicular heat management system when icing occurs in the outdoor heat exchanger in the heat pump mode.

(5) FIG. 5 is a flowchart showing an operation example of the vehicular heat management system according to the first embodiment of the present invention and showing an operation example depending on whether icing occurs in an outdoor heat exchanger under a heat pump mode condition.

(6) FIG. 6 is a view showing a vehicular heat management system according to a second embodiment of the present invention.

(7) FIG. 7 is a flowchart showing an operation example of the vehicular heat management system according to the second embodiment of the present invention and showing an operation example depending on whether icing occurs in an outdoor heat exchanger under a heat pump mode condition.

DETAILED DESCRIPTION

(8) Preferred embodiments of a vehicular heat management system according to the present invention will now be described in detail with reference to the accompanying drawings.

First Embodiment

(9) Prior to describing the features of the vehicular heat management system according to the present invention, the general configurations of the vehicular heat management system will be briefly described with reference to FIG. 2.

(10) The vehicular heat management system includes an air conditioner **100** that cools and heats the vehicle interior, and a water-cooled cooling device **200** that cools the battery B and the electric component module P.

(11) The air conditioner **100** is of a heat pump type and includes a refrigerant circulation line **110**.

(12) The refrigerant circulation line **110** includes a compressor **112**, a high-pressure side indoor heat exchanger **114**, a first expansion valve **116**, an outdoor heat exchanger **118**, a second expansion valve **120**, a third expansion valve **122** parallel to the second expansion valve **120**, a chiller **124** arranged on the downstream side of the second expansion valve **120**, and a low-pressure side indoor heat exchanger **126** on the downstream side of the third expansion valve **122**.

(13) The first expansion valve **116** and the second expansion valve **120** are of an electromagnetic type (EXV), and the third expansion valve **122** is of a thermostatic type (TXV) having an ON and OFF function.

(14) In the refrigerant circulation line **110**, the first expansion valve **116** is opened in an air conditioner mode, so that the refrigerant in the compressor **112** can circulate in an order of the high-pressure side indoor heat exchanger **114**, the outdoor heat exchanger **118**, the second expansion valve **120**, the third expansion valve **122**, the chiller **124**, and the low-pressure side indoor heat exchanger **126**.

(15) Through this refrigerant circulation, a low-temperature cold air is generated in the chiller **124** and the low-pressure indoor heat exchanger **126**. The generated cold air is delivered to the vehicle interior, the battery B, and the electric component module P. Therefore, the vehicle interior, the battery B, and the electric component module P are cooled.

(16) In addition, in a heat pump mode, the first expansion valve **116** is turned on, so that the refrigerant in the compressor **112** circulates in an order of the high-pressure side indoor heat exchanger **114**, the first expansion valve **116**, and the outdoor heat exchanger **118**.

(17) In addition, high-temperature heat is generated in the high-pressure side indoor heat exchanger **114** through the circulation of the refrigerant. The generated heat is supplied to the vehicle interior to heat the vehicle interior.

(18) Meanwhile, in the heat pump mode, the chiller **124** serves also to allow the refrigerant in the refrigerant circulation line **110** returning from the outdoor heat exchanger **118** to the compressor **112** to exchange heat with the cooling water on the side of a cooling water circulation line **210** of a water-cooled cooling device **200** that has absorbed the waste heat of the battery B and the electric

component module P.

(19) Therefore, in the heat pump mode, the waste heat from the battery B and the electric component module P can be recovered to the refrigerant in the refrigerant circulation line **110**.

(20) Further, the second expansion valve **120** on the upstream side of the chiller **124** is configured to variably adjust the depressurization/expansion amount of the refrigerant introduced into the chiller **124** even in the heat pump mode.

(21) Therefore, in the heat pump mode, it is possible to variably adjust the amount of heat absorbed by the chiller **124** and the outdoor heat exchanger **118** on the upstream side.

(22) In particular, the temperature of the outdoor heat exchanger **118** can be adjusted by allowing the heat absorption amount of the upstream-side outdoor heat exchanger **118** to be variably adjusted.

(23) Next, features of the vehicular heat management system according to the present invention will be described in detail with reference to FIGS. 2 to 5.

(24) Referring first to FIG. 2, the vehicular heat management system according to the present invention includes an icing occurrence determination part **300** that determines whether icing occurs in the outdoor heat exchanger **118** when entering the heat pump mode.

(25) The icing occurrence determination part **300** includes an outdoor air temperature sensor **310** that detects the temperature outside the vehicle interior, and a control part **320** that determines whether icing occurs in the outdoor heat exchanger **118** based on the outdoor air temperature data inputted from the outside temperature sensor **310**.

(26) The outdoor air temperature sensor **310** is installed on the outside of the vehicle interior, and is configured to detect the outdoor air temperature and input the detected outdoor air temperature data to the control part **320**.

(27) The control part **320** is provided with a microprocessor. When the outdoor air temperature data is inputted from the outdoor air temperature sensor **310** under the heat pump mode, the control part **320** determines whether the inputted outdoor air temperature falls within a temperature range in which icing occurs in the outdoor heat exchanger **118**, thereby determining whether icing occurs in the outdoor heat exchanger **118**.

(28) That is, as shown in Formula 1 below, it is determined whether the outdoor air temperature falls within a range exceeding a first set temperature and less than a second set temperature higher than the first set temperature.

First set temperature < Outdoor air temperature < Second set temperature Formula 1

(29) As a result of the determination, if the outdoor air temperature does not fall within the range exceeding the first set temperature and less than the second set temperature, the control part **320** determines that no icing has occurred in the outdoor heat exchanger **118**.

(30) As a result of the determination, if the outdoor air temperature falls within the range exceeding the first set temperature and less than the second set temperature, the control part **320** determines that icing has occurred in the outdoor heat exchanger **118**.

(31) In this regard, it is preferable that the first set temperature, which is a criterion for determining whether icing occurs in the outdoor heat exchanger **118**, is set to -5 degrees C., and the second set temperature is set to 5 degrees C.

(32) This is because it was found that icing occurs in the outdoor heat exchanger **118** when the outdoor air temperature is in the range between -5 degrees C. and 5 degrees C. under the heat pump mode condition.

(33) Referring again to FIG. 2, after determining whether icing has occurred in the outdoor heat exchanger **118**, the control part **320** controls the first expansion valve **116** and the second expansion valve **120** on the upstream and downstream sides of the outdoor heat exchanger **118** according to the determination result to prevent the occurrence of icing in the outdoor heat exchanger **118**.

(34) Explaining this in more detail, if it is determined that no icing has occurred in the outdoor heat exchanger **118**, the control part **320** enters the single expansion mode **320a**. The control part **320** is

configured to control only one of the first expansion valve **116** and the second expansion valve **120** in the single expansion mode **320a**.

(35) In particular, the second expansion valve **120** is fully opened, and only the opening degree is controlled for the first expansion valve **116** to adjust the depressurization/expansion amount of the refrigerant introduced into the outdoor heat exchanger **118**.

(36) Therefore, as shown in the p-h diagram of FIG. 3, it is possible to maintain a high refrigerant evaporation rate (D-D') in the outdoor heat exchanger **118**. Through this high refrigerant evaporation rate, the outdoor heat exchanger **118** makes it possible to control the heat generation amount of the high-pressure side indoor heat exchanger **114** while maintaining the temperature lower than the outdoor air temperature.

(37) When it is determined that icing has occurred in the outdoor heat exchanger **118**, the control part **320** enters a dual expansion mode **320b** as shown in FIG. 2. In the dual expansion mode **320b**, the control part **320** is configured to control both the first expansion valve **116** and the second expansion valve **120**.

(38) In particular, by variably controlling the opening degree of the second expansion valve **120**, the refrigerant on the downstream side of the outdoor heat exchanger **118** is once again decompressed and expanded (D'-D'') as shown in the p-h diagram of FIG. 4. The flow rate of the refrigerant on the side of the outdoor heat exchanger **118** is reduced during the process of depressurization and expansion of the refrigerant on the downstream side.

(39) Accordingly, the refrigerant evaporation (D-D') rate in the outdoor heat exchanger **118** is lowered, and the outdoor heat exchanger **118** can maintain a temperature higher than the outdoor temperature due to this low refrigerant evaporation rate.

(40) Thus, it is possible to cope with the occurrence of icing in the outdoor heat exchanger **118**. In particular, since the outdoor heat exchanger **118** is maintained at a higher temperature than the outdoor air temperature, the occurrence of icing in the outdoor heat exchanger **118** is fundamentally prevented. Even when icing occurs in the outdoor heat exchanger **118**, the icing can be removed quickly.

(41) Referring again to FIG. 2, the control part **320** controls the opening degree of the second expansion valve **120** in the state of entering the dual expansion mode **320b** to prevent the occurrence of icing in the outdoor heat exchanger **118**. Thereafter, the control part **320** periodically predicts the probability of occurrence of icing in the outdoor heat exchanger **118** according to a preset logic, and periodically variably controls the opening degree of at least one of the first and second expansion valves **116** and **120** according to the predicted probability of occurrence of icing in the outdoor heat exchanger **118**.

(42) That is, in the dual expansion mode **320b**, the control part **320** periodically calculates an icing occurrence index I in the outdoor heat exchanger **118** at predetermined time intervals through formula 2 below. For example, the icing occurrence index I is calculated at intervals of 30 seconds. Icing occurrence index(I)=Outdoor air temperature(Tamb)–Outdoor heat exchanger outlet refrigerant temperature(Tref) Formula 2

(43) When the calculation of the icing occurrence index I is completed, the control part **320** compares the calculated icing occurrence index I with a pre-stored reference index Tset to predict the probability of occurrence of icing in the outdoor heat exchanger **118**, and variably control the opening degree of the second expansion valve **120** according to the predicted probability of occurrence of icing.

(44) In particular, the control part **320** compares the icing occurrence index I with the reference index Tset, predicts the probability of occurrence of icing in the outdoor heat exchanger **118** according to the comparison result, and increases or decreases the opening degree of the second expansion valve **120** by a preset value according to the predicted probability of occurrence of icing.

(45) That is, for example, if it is confirmed that the calculated icing occurrence index I is greater than the reference index Tset, the control part **320** determines that the outdoor heat exchanger **118**

has a high probability of occurrence of icing in the outdoor heat exchanger **118**. According to this determination, the control part **320** reduces the opening degree of the second expansion valve **120** by a predetermined value.

(46) Therefore, the flow rate of the refrigerant in the outdoor heat exchanger **118** is reduced by a predetermined amount, thereby reducing the evaporation rate of the refrigerant in the outdoor heat exchanger **118** so that the temperature of the outdoor heat exchanger **118** can be maintained at a higher level.

(47) As a result, it is possible to fundamentally prevent the occurrence of icing in the outdoor heat exchanger **118** by actively coping with the high probability of occurrence of icing in the outdoor heat exchanger **118**.

(48) Conversely, if it is confirmed that the calculated icing occurrence index I is smaller than the reference index T_{set} , the control part **320** determines that the probability of occurrence of icing in the outdoor heat exchanger **118** is low. According to this determination, the control part **320** increases the opening degree of the second expansion valve **120** by a predetermined value.

(49) Therefore, the flow rate of the refrigerant in the outdoor heat exchanger **118** is increased by a predetermined amount, thereby increasing the evaporation rate of the refrigerant in the outdoor heat exchanger **118** so that the temperature of the outdoor heat exchanger **118** can be maintained at a lower level.

(50) As a result, it is possible to actively cope with the low probability of occurrence of icing in the outdoor heat exchanger **118**.

(51) Meanwhile, the control part **320** controls the opening degree of the second expansion valve **120** to be maintained at the current state when the calculated icing occurrence index I is equal to the reference index T_{set} .

(52) Referring again to FIG. 2, in the dual expansion mode **320b**, the control part **320** fixes the opening degree of the first expansion valve **116** when variably controlling the opening degree of the second expansion valve **120**.

(53) In particular, when the opening degree of the second expansion valve **120** is variably controlled according to the icing occurrence probability in the outdoor heat exchanger **118**, the control part **320** controls the opening degree of the first expansion valve **116** at an opening degree available before the entry into the dual expansion mode **320b** or a preset opening degree.

(54) Therefore, in the dual expansion mode **320b**, the opening degree of the first expansion valve **116** is fixed, the depressurization/expansion amount of the refrigerant introduced into the outdoor heat exchanger **118** is kept constant, and the heat absorption temperature of the outdoor heat exchanger **118** is controlled while controlling the refrigerant flow rate on the outdoor heat exchanger **118** side only by adjusting the opening degree of the second expansion valve **120**.

(55) Meanwhile, in the dual expansion mode **320b**, depending on the circumstances, the opening degree of the second expansion valve **120** may be fixed at a constant value, and the opening degree of the first expansion valve **116** may be variably controlled.

(56) In particular, the opening degree of the second expansion valve **120** may be fixed at a preset value, and the opening degree of the first expansion valve **116** may be variably controlled according to the probability of occurrence of icing in the outdoor heat exchanger **118**.

(57) Therefore, it is possible to control the refrigerant flow rate in the outdoor heat exchanger **118** and adjust the heat absorption temperature in the outdoor heat exchanger **118** only by adjusting the opening degree of the first expansion valve **116**.

(58) Referring again to FIG. 2, in the dual expansion mode **320b**, if the outdoor air temperature inputted by the outdoor temperature sensor **310** does not fall within the icing occurrence temperature range of the outdoor heat exchanger **118**, i.e., if the outdoor air temperature is a first set temperature (-5 degrees C.) or lower, or a second set temperature (5 degrees C.) or higher, the control part **320** determines that there is no possibility of occurrence of icing in the outdoor heat exchanger **118**.

(59) According to this determination, the control part **320** enters the single expansion mode **320a** while being released from the dual expansion mode **320b**. The control part **320** that has entered the single expansion mode **320a** fully opens the second expansion valve **120** and controls only the first expansion valve **116** to adjust the depressurization/expansion amount of the refrigerant introduced into the outdoor heat exchanger **118**.

(60) Meanwhile, in the single expansion mode **320a**, the control part **320** periodically predicts the probability of occurrence of icing in the outdoor heat exchanger **118** according to a preset logic.

(61) In particular, as described above, the control part **320** periodically calculates the icing occurrence index **I** of the outdoor heat exchanger **118** at predetermined time intervals through Formula 2, and compares the calculated icing occurrence index **I** with a pre-stored reference index **Tset** to predict the probability of occurrence of icing in the outdoor heat exchanger **118**.

(62) As a result of predicting the probability of occurrence of icing in the outdoor heat exchanger **118**, if it is determined that the probability of occurrence of icing in the outdoor heat exchanger **118** is high, the control part **320** controls the first and second expansion valves **116** and **120** while switching from the single expansion mode **320a** to the dual expansion mode **320b**.

(63) In particular, the control part **320** controls the first and second expansion valves **116** and **120** to induce expansion of the refrigerant twice in the outdoor heat exchanger **118**. By inducing the expansion of the refrigerant twice, the control part **320** adjusts the flow rate of the refrigerant in the outdoor heat exchanger **118**. By adjusting the flow rate of the refrigerant in the outdoor heat exchanger **118**, the control part **320** controls the temperature of the outdoor heat exchanger **118** and fundamentally prevents occurrence of a surface icing phenomenon.

(64) Next, an operation example of the vehicular heat management system according to the present invention having such a configuration will be described in detail with reference to FIGS. 2 and 5.

(65) Referring first to FIG. 5, in the heat pump mode state (**S101**), it is determined whether the outdoor air temperature falls within a range exceeding a first set temperature (−5 degrees C.) and less than a second set temperature (5 degrees C.) (**S103**).

(66) As a result of the determination, if the outdoor air temperature does not fall within the range exceeding the first set temperature (−5 degrees C.) and less than the second set temperature (5 degrees C.) (**S103-1**), the control part **320** determines that no icing has occurred in the outdoor heat exchanger **118**. According to this determination, the control part **320** enters a single expansion mode **320a** (**S105**).

(67) As shown in FIGS. 2 and 5, the control part **320** that has entered the single expansion mode **320a** fully opens the second expansion valve **120** and controls only the first expansion valve **116** to variably adjust the depressurization/expansion amount of the refrigerant introduced into the outdoor heat exchanger **118** (**S107**).

(68) Meanwhile, as a result of the determination in step **S103**, if the outdoor air temperature falls within the range exceeding the first set temperature (−5 degrees C.) and less than the second set temperature (5 degrees C.) (**S103-2**), the control part **320** determines that icing has occurred in the outdoor heat exchanger **118**. According to this determination, the control part **320** enters a dual expansion mode **320b** (**S109**).

(69) The control part **320** that has entered the dual expansion mode **320b** controls the opening degrees of the first expansion valve **116** and the second expansion valve **120** (**S111**).

(70) In particular, by variably controlling the opening degree of the second expansion valve **120**, the control part **320** depressurizes and expands the refrigerant on the downstream side of the outdoor heat exchanger **118** once again, thereby reducing the refrigerant flow rate in the outdoor heat exchanger **118** and lowering the refrigerant evaporation rate in the outdoor heat exchanger **118**.

(71) Then, the temperature of the outdoor heat exchanger **118** is increased so as to prevent occurrence of icing on the surface and remove the generated icing.

(72) Meanwhile, in the dual expansion mode **320b**, the control part **320** periodically calculates the

icing occurrence index I of the outdoor heat exchanger **118** through Formula 2 at predetermined time intervals (S113).

(73) When the calculation of the icing occurrence index I is completed, the control part **320** periodically determines whether the calculated icing occurrence index I is greater than a pre-stored reference index Tset (S115).

(74) As a result of the determination, if the icing occurrence index I is greater than the reference index Tset, the control part **320** determines that the probability of occurrence of icing in the outdoor heat exchanger **118** is high, thereby reducing the opening degree of the second expansion valve **120** by a preset value (S117).

(75) Then, the flow rate of the refrigerant in the outdoor heat exchanger **118** decreases, and the evaporation rate of the refrigerant in the outdoor heat exchanger **118** decreases. The temperature of the outdoor heat exchanger **118** increases due to the reduced evaporation rate. This makes it possible to cope with the probability of occurrence of icing in the outdoor heat exchanger **118**.

(76) Meanwhile, as a result of the determination in step S115, if the icing occurrence index I is not greater than the reference index Tset (S115-1), the control part **320** determines again whether the icing occurrence index I is smaller than the reference index Tset (S119).

(77) As a result of the determination, if the icing generation index I is smaller than the reference index Tset, the control part **320** determines that the probability of occurrence of icing in the outdoor heat exchanger **118** is low, thereby increasing the opening degree of the second expansion valve **120** by a preset value (S121).

(78) Then, the flow rate of the refrigerant in the outdoor heat exchanger **118** increases, and the evaporation rate of the refrigerant in the outdoor heat exchanger **118** increases. The temperature of the outdoor heat exchanger **118** is decreased due to the increased evaporation rate. This makes it possible to cope with the probability of occurrence of icing in the outdoor heat exchanger **118**.

(79) Meanwhile, as a result of the determination in step S119, if the icing occurrence index I is not smaller than the reference index Tset (S119-1), the control part **320** determines again whether the icing occurrence index I is equal to the reference index Tset (S123).

(80) As a result of the determination, if the icing occurrence index I is equal to the reference index Tset, the control part **320** keeps the opening degree of the second expansion valve **120** at the current state (S125).

(81) According to the vehicular heat management system of the present invention, when icing occurs in the outdoor heat exchanger **118** under the heat pump mode condition, the temperature of the outdoor heat exchanger **118** is adjusted by controlling the opening degree of the second expansion valve **120** on the downstream side of the outdoor heat exchanger **118**. Therefore, when icing occurs in the outdoor heat exchanger **118**, it is possible to cope with the occurrence of icing in the outdoor heat exchanger **118** without restricting the flow of the refrigerant to the outdoor heat exchanger **118**.

(82) In addition, since it is possible to cope with the occurrence of icing in the outdoor heat exchanger **118** without restricting the flow of the refrigerant to the outdoor heat exchanger **118**, the unstable flow of the refrigerant in the refrigerant circulation line due to the restriction of the flow of the refrigerant to the outdoor heat exchanger **118** can be prevented when icing occurs in the outdoor heat exchanger **118**.

(83) In addition, since the instability of the flow of the refrigerant in the refrigerant circulation line **110** due to the restriction of the flow of the refrigerant to the outdoor heat exchanger **118** can be prevented, it is possible to prevent the fluctuation of the heat exchange rate in the high-pressure side indoor heat exchanger **114** due to the instability of the flow of the refrigerant in the refrigerant circulation line **110** and the resultant change in the temperature of the air discharged into the vehicle interior, thereby improving the heating performance in the vehicle interior.

(84) In addition, since it is possible to cope with the occurrence of icing in the outdoor heat exchanger **118** without restricting the flow of the refrigerant to the outdoor heat exchanger **118**

when icing occurs in the outdoor heat exchanger **118**, the decrease in the refrigerant heat exchange efficiency (evaporation efficiency) on the outdoor heat exchanger **118** side due to the restriction of the refrigerant flow to the outdoor heat exchanger **118** can be prevented.

(85) In addition, since it is possible to prevent the decrease in the refrigerant heat exchange efficiency (evaporation efficiency) on the outdoor heat exchanger **118** side due to the restriction of the refrigerant flow to the outdoor heat exchanger **118**, it is possible to prevent a decrease in the heat generation amount in the high-pressure side indoor heat exchanger **114**, thereby improving the heating performance in the vehicle interior.

Second Embodiment

(86) FIG. **6** is a diagram showing a vehicular heat management system according to a second embodiment of the present invention.

(87) The vehicular heat management system according to the second embodiment has the same main configurations as those of the vehicular heat management system according to the first embodiment described above.

(88) In particular, the vehicular heat management system according to the second embodiment has the same structure as the vehicular heat management system according to the first embodiment in that when icing occurs in the outdoor heat exchanger **118** under the heat pump mode condition, the temperature of the outdoor heat exchanger **118** is adjusted by controlling the opening degree of the second expansion valve **120** on the downstream side of the outdoor heat exchanger **118**, whereby the occurrence of icing in the outdoor heat exchanger **118** can be prevented without restricting the flow of the refrigerant to the outdoor heat exchanger **118** when icing occurs in the outdoor heat exchanger **118**.

(89) However, the vehicular heat management system according to the second embodiment differs from the vehicular heat management system according to the first embodiment in terms of the structure of the icing occurrence determination part **300** for determining whether icing occurs in the outdoor heat exchanger **118**.

(90) In particular, the icing occurrence determination part **300** of the first embodiment is configured to determine whether icing occurs in the outdoor heat exchanger **118** based on the outdoor air temperature data. However, the icing occurrence determination part **300** of the second embodiment is configured to determine whether icing occurs in the outdoor heat exchanger **118** based on two types of data, i.e., the outdoor air temperature and the outdoor air humidity.

(91) Explaining this in more detail, the icing occurrence determination part **300** of the second embodiment further includes an outdoor air humidity sensor **330**.

(92) The outdoor air humidity sensor **330** is installed outside the vehicle interior, and is configured to detect the relative humidity of the outdoor air (hereinafter abbreviated as “outdoor air humidity”) and input the detected outdoor air humidity data to the control part **320**.

(93) Meanwhile, when the outdoor air temperature data and the outdoor air humidity data are respectively inputted from the outdoor air temperature sensor **310** and the outdoor air humidity sensor **330** in the heat pump mode, the control part **320** determines whether the outdoor air temperature and the outdoor air humidity satisfy all the conditions set in advance, respectively, and then determines whether icing occurs in the outdoor heat exchanger **118** depending on whether all of the conditions are satisfied.

(94) That is, when the outdoor air temperature data and the outdoor air humidity data are respectively inputted from the outdoor air temperature sensor **310** and the outdoor air humidity sensor **330**, respectively, the control part **320** determines whether the inputted outdoor air temperature satisfies a condition (hereinafter referred to as “first condition”) in which the outdoor air temperature falls within a range exceeding a first set temperature and less than a second set temperature as in Formula 1 described above.

(95) Then, the control part **320** determines whether the inputted outdoor air humidity satisfies a condition (hereinafter referred to as “second condition”) in which the outdoor air humidity is equal

to or greater than a preset reference humidity as shown in Formula 3 below.

Outdoor air humidity $\geq$ Reference humidity Formula 3

(96) As a result of the determination, if either of the first condition and the second condition is not satisfied, that is, if either of the first condition in which the outdoor air temperature falls within a range exceeding a first set temperature and less than a second set temperature and the second condition in which the outdoor air humidity is equal to or greater than a preset reference humidity is not satisfied, the control part **320** determines that no icing has occurred in the outdoor heat exchanger **118**.

(97) If all of the first condition and the second condition are satisfied, that is, if all of the first condition in which the outdoor air temperature falls within a range exceeding a first set temperature and less than a second set temperature and the second condition in which the outdoor air humidity is equal to or greater than a preset reference humidity are satisfied, the control part **320** determines that icing has occurred in the outdoor heat exchanger **118**.

(98) According to the determination results, as in the first embodiment, the control part **320** controls the first and second expansion valves **116** and **120** while entering the single expansion mode **320a** or the dual expansion mode **320b**.

(99) In this regard, the reference humidity, which is a criterion for determining whether icing occurs in the outdoor heat exchanger **118**, is preferably set to 75% relative humidity.

(100) This is because icing occurs in the outdoor heat exchanger **118** when the outdoor air humidity is equal to or greater than the 75% relative humidity under the heat pump mode condition.

(101) Next, an operation example of the vehicular heat management system according to the second embodiment having such a configuration will be described in detail with reference to FIGS. **6** and **7**.

(102) Referring first to FIG. **7**, in the heat pump mode (**S201**), the control part **320** determines whether all of a condition in which the outdoor air temperature falls within a range exceeding a first set temperature (−5 degrees C.) and less than a second set temperature (5 degrees C.) and a condition in which the outdoor air humidity is equal to or greater than a reference humidity (75%) are satisfied (**S203**).

(103) As a result of the determination, if both the condition in which the outdoor air temperature falls within the range exceeding the first set temperature (−5 degrees C.) and less than the second set temperature (5 degrees C.) and the condition in which the outdoor air humidity is equal to or greater than the reference humidity (75%) are not satisfied (**S203-1**), the control part **320** determines that no icing has occurred in the outdoor heat exchanger **118**, and enters the single expansion mode **320a** according to this determination (**S205**).

(104) An operation example after entering the single expansion mode **320a** is the same as that of the first embodiment described above. Therefore, a description thereof will be omitted.

(105) On the other hand, as a result of the determination in step **S203**, if both the condition in which the outdoor air temperature falls within the range exceeding the first set temperature (−5 degrees C.) and less than the second set temperature (5 degrees C.) and the condition in which the outdoor air humidity is equal to or greater than the reference humidity (75%) are satisfied (**S203-2**), the control part **320** determines that icing has occurred in the outdoor heat exchanger **118**, and enters the dual expansion mode **320b** according to this determination (**S209**).

(106) An operation example after entering the dual expansion mode **320b** is the same as that of the first embodiment described above. Therefore, a description thereof will be omitted.

(107) According to the vehicular heat management system according to the second embodiment having such a configuration, the control part **320** is configured to determine whether icing occurs in the outdoor heat exchanger **118** based on the outdoor air temperature data and the outdoor air humidity data. This makes it possible to accurately determine whether icing occurs in the outdoor heat exchanger **118**.

(108) In addition, since it is possible to accurately determine whether icing occurs in the outdoor

heat exchanger **118**, the time point of entry into the dual expansion mode **320b** due to the occurrence of icing in the outdoor heat exchanger **118** can be precisely controlled.

(109) In particular, since it is possible to precisely control the time point of entry into the dual expansion mode **320b**, the unstable refrigerant flow in the refrigerant circulation line **110** due to the entry into the dual expansion mode **320b** can be minimized. This makes it possible to improve the performance of the pump mode.

(110) While the preferred embodiments of the present invention have been described above, the present invention is not limited to the above-described embodiments. Various modifications and changes may be made without departing from the scope and spirit of the present invention defined in the claims.

Claims

1. A vehicular heat management system, comprising: a compressor arranged on a heat pump type refrigerant circulation line; a high-pressure side heat exchanger arranged on the heat pump type refrigerant circulation line; an outdoor heat exchanger arranged on the heat pump type refrigerant circulation line; a low-pressure side heat exchanger arranged on the heat pump type refrigerant circulation line; a first expansion valve arranged on the upstream side of the outdoor heat exchanger; a second expansion valve arranged on the downstream side of the outdoor heat exchanger; and a control part configured to control opening degrees of the first expansion valve and the second expansion valve depending on whether icing occurs in the outdoor heat exchanger under a heat pump mode condition, wherein the control part is configured to enter a dual expansion mode in which a refrigerant is expanded by the first expansion valve and the second expansion valve if the control part determines that icing occurs in the outdoor heat exchanger, and enter a single expansion mode in which a refrigerant is expanded by only one of the first expansion valve and the second expansion valve if the control part determines that icing does not occur in the outdoor heat exchanger, wherein the control part is configured to, in the single expansion mode, fully open the second expansion valve on the downstream side of the outdoor heat exchanger and control the opening degree of the first expansion valve on the upstream side of the outdoor heat exchanger to adjust a refrigerant flow rate on the side of the outdoor heat exchanger, and wherein the control part is configured to, in the dual expansion mode, control the opening degrees of the first expansion valve and the second expansion valve to adjust the refrigerant flow rate on the side of the outdoor heat exchanger in a decreasing direction so that the temperature of the outdoor heat exchanger is increased to cope with occurrence of icing.

2. The system of claim 1, wherein the control part is configured to determine that icing occurs in the outdoor heat exchanger when an outdoor air temperature falls within a range exceeding a first set temperature and less than a second set temperature higher than the first set temperature, and the control part is configured to determine that icing does not occur in the outdoor heat exchanger when the outdoor air temperature does not fall within the range exceeding the first set temperature and less than the second set temperature.

3. The system of claim 1, wherein the control part is configured to determine that icing has occurred in the outdoor heat exchanger when both a first condition in which an outdoor air temperature falls within a range exceeding a first set temperature and less than a second set temperature and a second condition in which an outdoor air humidity is equal to or greater than a preset reference humidity are satisfied, and the control part determines that no icing has occurred in the outdoor heat exchanger when either of the first condition and the second condition is not satisfied.

4. A vehicular heat management system, comprising: a compressor arranged on a heat pump type refrigerant circulation line; a high-pressure side heat exchanger arranged on the heat pump type refrigerant circulation line; an outdoor heat exchanger arranged on the heat pump type refrigerant

circulation line; a low-pressure side heat exchanger arranged on the heat pump type refrigerant circulation line; a first expansion valve arranged on the upstream side of the outdoor heat exchanger; a second expansion valve arranged on the downstream side of the outdoor heat exchanger; and a control part configured to control opening degrees of the first expansion valve and the second expansion valve depending on whether icing occurs in the outdoor heat exchanger under a heat pump mode condition, wherein the control part is configured to enter a dual expansion mode in which a refrigerant is expanded by the first expansion valve and the second expansion valve if the control part determines that icing occurs in the outdoor heat exchanger, and enter a single expansion mode in which a refrigerant is expanded by only one of the first expansion valve and the second expansion valve if the control part determines that icing does not occur in the outdoor heat exchanger, wherein the control part is configured to, in the dual expansion mode, keep the opening degree of one of the first expansion valve and the second expansion valve constant, and variably control the opening degree of the other of the first expansion valve and the second expansion valve, and wherein the control part is configured to, in the dual expansion mode, keep the opening degree of one of the first expansion valve and the second expansion valve constant, and variably control the opening degree of the other of the first expansion valve and the second expansion valve.

5. The system of claim 4, wherein the control part is configured to, in the dual expansion mode, periodically predict the probability of occurrence of icing in the outdoor heat exchanger according to a pre-stored logic, and variably control the opening degree of the second expansion valve according to the predicted probability of occurrence of icing.

6. The system of claim 5, wherein the control part is configured to, in the dual expansion mode, periodically calculate an icing occurrence index of the outdoor heat exchanger at predetermined time intervals through pre-stored Formula: icing occurrence index (I)=outdoor air temperature (Tamb)–outdoor heat exchanger outlet refrigerant temperature (Tref), the control part is configured to predict that the probability of occurrence of icing in the outdoor heat exchanger is high if the calculated icing occurrence index is greater than a pre-stored reference index, and the control part is configured to predict that the probability of occurrence of icing in the outdoor heat exchanger is low if the calculated icing occurrence index is smaller than the reference index.

7. The system of claim 6, wherein the control part is configured to, when it is determined that the probability of occurrence of icing in the outdoor heat exchanger is high, reduce the opening degree of the second expansion valve by a preset value to reduce the refrigerant flow rate in the outdoor heat exchanger by a preset value.

8. The system of claim 7, wherein the control part is configured to, when it is determined that the probability of occurrence of icing in the outdoor heat exchanger is low, increase the opening degree of the second expansion valve by a preset value to increase the refrigerant flow rate in the outdoor heat exchanger by a preset value.

9. The system of claim 8, wherein the control part is configured to keep the opening degree of the second expansion valve at a current state when the icing occurrence index is equal to the reference index.
